

At the beginning of this course, I understood telecommunications as simply the process of sending information from one place to another through phone calls, text messages, or the internet. As I completed the assignments and laboratory exercises, I developed a much deeper understanding of how communication systems actually work. I explored essential technical components and various structural frameworks that define overall system functionality mostly around transmitters, channels and receivers.

I also learned about more advanced telecommunications concepts such as 5G Core, network slicing, Multi-access Edge Computing and the Service-Based Architecture. I can now understand how these technologies work together to provide faster communication and lower latency for different applications like in healthcare or autonomous systems.

In the laboratory 4 assignment, the exercises helped connect the theory to practical applications. Working with simulated network traffic showed me how communication networks experience predictable daily and weekly usage patterns and how these patterns can be analyzed to improve network performance.

Overall, this course changed my view of telecommunications from simply using communication technology to understanding the intelligent systems that make modern communication possible.